

Positive coordinate-decomposition cases of the countable-chain-condition conjecture in free quasi-Banach lattices

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Status

This is an expanded partial solution packet for the open question in Salguero-Alarcon, Tradacete, and Trejo-Arroyo, *Free quasi-Banach lattices*, arXiv:2512.05273. The claims below give positive coordinate-decomposition families for their countable-chain-condition conjecture, subject to human review.

Source problem

The source paper fixes $0 < p \leq 1$, defines

$$E^\Delta = \mathcal{L}(E, L_p[0, 1]),$$

and gives B_{E^Δ} the weak- Δ topology $\sigma(E^\Delta, E)$, the initial topology for all evaluation maps $T \mapsto T(e) \in L_p[0, 1]$. The paper proves that

$$C_{\text{ph}}^b(B_{\ell_p(\Gamma)^\Delta}, L_p[0, 1])$$

has the countable chain condition for every set Γ , and then asks whether the same conclusion holds for every p -natural quasi-Banach space E . The relevant source passage is reproduced in Figure 1.

7.2. $\text{FpBL}^{(p)}[E]$ and the countable chain condition. We conjecture that Theorem 6.10 is true for *any* p -natural quasi-Banach space E ; that is, $C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$ satisfies the countable chain condition regardless of E . The arguments of Section 6 can be carefully adapted to show that this is certainly the case when E is separable. Indeed, since there is a quotient map $\pi : \ell_p \rightarrow E$, then the map $\pi^\Delta : E^\Delta \hookrightarrow \ell_p^\Delta$ is $\sigma(E^\Delta, E)$ - $\sigma(\ell_p^\Delta, \ell_p)$ -continuous. Hence, B_{E^Δ} is a subspace of the separable and metrizable space $B_{\ell_p^\Delta} = (B_{L_p[0, 1]})^\mathbb{N}$ and therefore, B_{E^Δ} has the (topological) countable chain condition, which is enough to conclude. However, since the topological countable chain condition is not hereditary, this argument does not seem to admit a straightforward generalization.

Figure 1: Source-paper Section 7.2, stating the CCC conjecture.

Transcription of the open question

Section 7.2 of the source paper states:

We conjecture that Theorem 6.10 is true for any p -natural quasi-Banach space E ; that is,

$$C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$$

satisfies the countable chain condition regardless of E .

The paper observes that the separable case follows because B_{E^Δ} is then a subspace of the separable metrizable space $(B_{L_p[0,1]})^\mathbb{N}$, while topological CCC is not hereditary in general.

Definitions

A vector lattice X satisfies the countable chain condition, abbreviated CCC, if every pairwise disjoint family in X_+ is countable. A topological space S satisfies topological CCC if every pairwise disjoint family of nonempty open subsets of S is countable.

For a family $(E_\gamma)_{\gamma \in \Gamma}$ of p -Banach spaces and $p \leq r < \infty$,

$$\left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_r$$

denotes the usual ℓ_r -sum: $x = (x_\gamma)$ belongs to it when $\sum_\gamma \|x_\gamma\|^r < \infty$, with $\|x\|^r = \sum_\gamma \|x_\gamma\|^r$. We also use

$$\left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{c_0}$$

for the c_0 -sum, with norm $\|x\| = \sup_\gamma \|x_\gamma\|$ and the usual vanishing-at-infinity condition.

We shall also use the following abstract form of a coordinate decomposition. A p -Banach space E has a *bounded separable Boolean coordinate decomposition* over a set Γ if, for every $A \subseteq \Gamma$, there is a projection $P_A : E \rightarrow E$ such that:

$$P_\emptyset = 0, \quad P_\Gamma = I, \quad P_A P_B = P_{A \cap B},$$

and $P_{A \cup B} = P_A + P_B$ whenever $A \cap B = \emptyset$; for every $x \in E$, $P_F x \rightarrow x$ as F increases through the finite subsets of Γ ; every coordinate range $P_{\{\gamma\}} E$ is separable; and

$$K := \sup_{A \subseteq \Gamma} \|P_A\| < \infty.$$

Proof intuition

The source proof for $\ell_p(\Gamma)$ works because

$$B_{\ell_p(\Gamma)^\Delta} \cong (B_{L_p[0,1]})^\Gamma,$$

and the right-hand side is a product of separable metric spaces, hence has topological CCC. The same product mechanism survives if each scalar coordinate is replaced by a separable p -natural

block. Operators from an ℓ_p -sum into L_p are exactly uniformly contractive families of operators on the summands: the p -triangle inequality in L_p makes the coordinatewise series converge with the expected norm bound.

Thus the weak- Δ unit ball of the whole sum is not merely embedded in a product; it is homeomorphic to the product of the block unit balls. The block unit balls are separable metrizable, so the product is topologically CCC. A general lifting lemma then turns topological CCC of the domain into lattice CCC for L_p -valued continuous positive functions.

There is a second, more flexible mechanism for spaces with a bounded separable Boolean coordinate decomposition. Their weak- Δ unit balls need not be topologically CCC; $c_0(\Gamma)$ already has weak-star dual ball $B_{\ell_1(\Gamma)}$, which has uncountably many disjoint weak-star slices. But those slices use thresholds and are not homogeneous. Homogeneous disjoint functions instead give disjoint relatively open cones. Finite-block operators are weak- Δ -dense; after shrinking them toward zero, a Delta-system and root-gluing argument shows that uncountably many relatively open cones cannot be pairwise disjoint. The proof uses only boundedness of the Boolean projections and separability of finite coordinate ranges.

Lemma 1 (Topological CCC lifts to the function lattice). *Let S be a topological space satisfying topological CCC. Then every sublattice of $C(S, L_p[0, 1])$ consisting of continuous $L_p[0, 1]$ -valued functions satisfies CCC. In particular,*

$$C_{\text{ph}}^b(S, L_p[0, 1])$$

satisfies CCC whenever it is defined.

Proof. It suffices to rule out an uncountable family $(f_i)_{i \in I}$ of nonzero positive pairwise disjoint continuous functions $S \rightarrow L_p[0, 1]$. Use the countable family $\mathcal{B}$ of open balls in $L_p[0, 1]$ from Lemma 6.9 of the source paper: it covers $L_p[0, 1]_+ \setminus \{0\}$, and whenever $u, v \in B \in \mathcal{B}$, one has $u \wedge v \neq 0$.

For each i , choose $s_i \in S$ with $f_i(s_i) \neq 0$, and then choose $B_i \in \mathcal{B}$ with $f_i(s_i) \in B_i$. Since $\mathcal{B}$ is countable, an uncountable subfamily has a common ball B . For that subfamily, the open sets

$$U_i = f_i^{-1}(B)$$

are nonempty. They are pairwise disjoint: if $s \in U_i \cap U_j$, then $f_i(s), f_j(s) \in B$, hence $f_i(s) \wedge f_j(s) \neq 0$, contradicting the disjointness $f_i \wedge f_j = 0$. This gives an uncountable pairwise disjoint family of nonempty open subsets of S , contradicting topological CCC. $\square$

Lemma 2 (A countable cone cover of L_{p+}). *There is a countable family $\mathcal{V}$ of open cones in $L_p[0, 1]_+ \setminus \{0\}$ such that $\mathcal{V}$ covers $L_p[0, 1]_+ \setminus \{0\}$, and whenever $u, v \in V \in \mathcal{V}$, one has $u \wedge v \neq 0$.*

Proof. Let $S_+ = \{u \in L_p[0, 1]_+ : \|u\|_p = 1\}$. Since $L_p[0, 1]$ is separable, so is S_+ ; choose a countable $1/3$ -net (b_n) in S_+ . Put

$$V_n = \left\{ u \in L_p[0, 1]_+ \setminus \{0\} : \left\| \frac{u}{\|u\|_p} - b_n \right\|_p < \frac{1}{3} \right\}.$$

These sets are open cones and cover $L_p[0, 1]_+ \setminus \{0\}$. If $u, v \in V_n$ and $u \wedge v = 0$, then their normalizations are disjoint positive unit vectors, so

$$\left\| \frac{u}{\|u\|_p} - \frac{v}{\|v\|_p} \right\|_p^p = 2.$$

On the other hand the p -triangle inequality gives this p -th power as less than $2(1/3)^p < 2$, a contradiction. $\square$

Lemma 3 (The weak- Δ ball of a p -sum). *Let $0 < p \leq 1$, and let $E = (\bigoplus_{\gamma \in \Gamma} E_\gamma)_p$. Then the restriction map*

$$\Phi : B_{E^\Delta} \longrightarrow \prod_{\gamma \in \Gamma} B_{E_\gamma^\Delta}, \quad \Phi(T) = (T \circ j_\gamma)_{\gamma \in \Gamma},$$

where $j_\gamma : E_\gamma \rightarrow E$ is the coordinate embedding, is a homeomorphism when all unit balls are given their weak- Δ topologies.

Proof. The map is well-defined and injective. To prove surjectivity, take a family $(T_\gamma)_\gamma \in \prod_\gamma B_{E_\gamma^\Delta}$. For $x = (x_\gamma) \in E$, the support of x is countable, and the net of finite partial sums

$$\sum_{\gamma \in F} T_\gamma x_\gamma$$

is Cauchy in $L_p[0, 1]$, since for finite $F_1 \subseteq F_2$,

$$\left\| \sum_{\gamma \in F_2 \setminus F_1} T_\gamma x_\gamma \right\|_p^p \leq \sum_{\gamma \in F_2 \setminus F_1} \|T_\gamma x_\gamma\|_p^p \leq \sum_{\gamma \in F_2 \setminus F_1} \|x_\gamma\|^p.$$

Define $Tx = \sum_\gamma T_\gamma x_\gamma$. The same estimate gives

$$\|Tx\|_p^p \leq \sum_\gamma \|x_\gamma\|^p = \|x\|^p,$$

so $T \in B_{E^\Delta}$, and $T \circ j_\gamma = T_\gamma$.

Continuity of Φ is immediate, since each coordinate $T \mapsto T \circ j_\gamma$ is determined by evaluations at vectors of E . For continuity of the inverse, fix $x = (x_\gamma) \in E$. We must show that

$$(T_\gamma)_\gamma \longmapsto \sum_\gamma T_\gamma x_\gamma$$

is continuous from the product of weak- Δ topologies to the norm topology of $L_p[0, 1]$. Use the equivalent metric $d(u, v) = \|u - v\|_p^p$ on $L_p[0, 1]$. Given $\eta > 0$, choose a finite $F \subset \Gamma$ such that

$$\sum_{\gamma \notin F} \|x_\gamma\|^p < \eta/3.$$

The finite-coordinate map

$$(T_\gamma)_\gamma \longmapsto \sum_{\gamma \in F} T_\gamma x_\gamma$$

is continuous. Thus, near a fixed point $(T_\gamma)_\gamma$, one may force

$$\left\| \sum_{\gamma \in F} (S_\gamma - T_\gamma) x_\gamma \right\|_p^p < \eta/3.$$

For such $(S_\gamma)_\gamma$, the p -triangle inequality gives

$$\begin{aligned} \left\| \sum_\gamma (S_\gamma - T_\gamma) x_\gamma \right\|_p^p &\leq \left\| \sum_{\gamma \in F} (S_\gamma - T_\gamma) x_\gamma \right\|_p^p \\ &\quad + \left\| \sum_{\gamma \notin F} S_\gamma x_\gamma \right\|_p^p + \left\| \sum_{\gamma \notin F} T_\gamma x_\gamma \right\|_p^p \\ &< \eta. \end{aligned}$$

This proves continuity of Φ^{-1} . $\square$

Lemma 4 (Separable blocks have separable weak- Δ balls). *If E is separable, then B_{E^Δ} is separable metrizable in the weak- Δ topology.*

Proof. Let $D = \{d_n : n \in \mathbb{N}\}$ be dense in E . On the unit ball B_{E^Δ} , the topology generated by the countable family of evaluations $T \mapsto T(d_n)$ agrees with $\sigma(E^\Delta, E)$. Indeed, if $d_n \rightarrow e$, then uniformly over $T \in B_{E^\Delta}$,

$$\|T(d_n) - T(e)\|_p \leq \|d_n - e\|.$$

Thus every evaluation at $e \in E$ is continuous for the countably generated topology. Consequently B_{E^Δ} embeds topologically into the countable product $(L_p[0, 1])^\mathbb{N}$, a separable metrizable space, and so B_{E^Δ} is itself separable metrizable. $\square$

Theorem 1. *Let $0 < p \leq 1$, and let $(E_\gamma)_{\gamma \in \Gamma}$ be any family of separable p -natural quasi-Banach spaces. Put*

$$E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_p.$$

Then E is p -natural and

$$C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$$

satisfies the countable chain condition.

Proof. First, E is p -natural. For each γ , choose an isometric embedding of E_γ into a p -convex p -Banach lattice X_γ with $M^{(p)}(X_\gamma) = 1$. Then $(\bigoplus_\gamma X_\gamma)_p$ is again a p -convex p -Banach lattice with p -convexity constant 1, and the coordinatewise embedding of E is isometric.

By Lemma 4, each $B_{E_\gamma^\Delta}$ is separable metrizable. By the standard product theorem used in the source paper [2, 2.7.10(d)], arbitrary products of separable metrizable spaces have topological CCC. Hence

$$\prod_{\gamma \in \Gamma} B_{E_\gamma^\Delta}$$

has topological CCC. Lemma 3 identifies this product with B_{E^Δ} . Therefore B_{E^Δ} has topological CCC. Finally Lemma 1 implies that $C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$ has lattice CCC. $\square$

Corollary 1. *The source conjecture holds for all p -sums of separable p -natural blocks. This simultaneously recovers the source paper's separable case and its case $E = \ell_p(\Gamma)$.*

Proof. If Γ is a singleton, Theorem 1 is the separable case. If $E_\gamma = \mathbb{R}$ for every γ , then $(\bigoplus_\gamma E_\gamma)_p = \ell_p(\Gamma)$. $\square$

Theorem 2 (The coordinate-decomposition case). *Let $0 < p \leq 1$, and let E be a p -natural quasi-Banach space with a bounded separable Boolean coordinate decomposition $(P_A)_{A \subseteq \Gamma}$. Then*

$$C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$$

satisfies the countable chain condition.

Proof. Let $K = \sup_{A \subseteq \Gamma} \|P_A\|$. Assume, toward a contradiction, that $(f_i)_{i \in I}$ is an uncountable family of nonzero positive pairwise disjoint elements of $C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$. By Lemma 2, after passing to an uncountable subfamily there is a fixed open cone $V \subset L_p[0, 1]_+ \setminus \{0\}$ such that

$$U_i = f_i^{-1}(V)$$

is nonempty for every i . The sets U_i are relatively open cones in B_{E^Δ} , and they are pairwise disjoint.

We shall contradict this. Let $P_A : E \rightarrow E$ denote the coordinate projection onto a finite set $A \subset \Gamma$. For each $T \in B_{E^\Delta}$, $TP_A \rightarrow T$ pointwise on E as A increases through finite subsets of Γ , by hypothesis. Since the projections are only assumed bounded, we apply this to T/K : the operators $K^{-1}TP_A$ belong to B_{E^Δ} and converge pointwise to $K^{-1}T$. Since each U_i is a cone, this gives finite-block points of $U_i \cap B_{E^\Delta}$. Thus we may choose $S_i \in U_i$ with finite block support A_i , meaning $S_i = S_i P_{A_i}$, and $\|S_i\| < \eta$, where $\eta > 0$ is fixed so small that $(4(K\eta)^p)^{1/p} < 1$.

For each i , choose a basic weak- Δ neighborhood

$$N_i = \{T \in B_{E^\Delta} : \|Tx_{i,k} - S_i x_{i,k}\|_p < \varepsilon_i, \ 1 \leq k \leq m_i\}$$

with $N_i \subset U_i$. Enlarge A_i to a finite set H_i so that

$$\|P_{\Gamma \setminus H_i} x_{i,k}\| < \varepsilon_i / (8\eta K^2) \quad (1 \leq k \leq m_i).$$

Replacing A_i by H_i , apply the Delta-system lemma to obtain an uncountable subfamily, still indexed by I , for which the finite sets (A_i) have a common root R and pairwise disjoint tails $A_i \setminus R$.

Let $E_R = P_R E$. This space is separable because R is finite and each $P_{\{\gamma\}} E$ is separable. The set

$$\mathcal{R} = \{R_0 \in \mathcal{L}(E_R, L_p[0, 1]) : \|R_0\| < 2\eta\},$$

with the topology of pointwise convergence on E_R , is separable metrizable. For each i , let $O_i \subset \mathcal{R}$ be the set of all root operators such that, if S_i is replaced by R_0 on the root coordinates and left unchanged on $A_i \setminus R$, then the inequalities defining N_i hold with margin $\varepsilon_i/2$. Each O_i is a nonempty relatively open subset of $\mathcal{R}$, and $S_i|_{E_R} \in O_i$. Therefore two of them, say O_i and O_j , intersect. Pick $R_0 \in O_i \cap O_j$.

Define $S \in E^\Delta$ blockwise as follows: on the root R , let $S = R_0 P_R$; on $A_i \setminus R$, let $S = S_i P_{A_i \setminus R}$; on $A_j \setminus R$, let $S = S_j P_{A_j \setminus R}$; and on all remaining coordinates let $S = 0$. In other words,

$$S = R_0 P_R + S_i P_{A_i \setminus R} + S_j P_{A_j \setminus R}.$$

Since the three displayed pieces are supported on disjoint coordinate sets and all coordinate projections have norm at most K , for $\|x\| \leq 1$ the p -triangle inequality gives

$$\|Sx\|_p^p \leq \|R_0 P_R x\|_p^p + \|S_i P_{A_i \setminus R} x\|_p^p + \|S_j P_{A_j \setminus R} x\|_p^p \leq (2^p + 2)(K\eta)^p < 4(K\eta)^p < 1,$$

so $S \in B_{E^\Delta}$.

We claim that $S \in N_i \cap N_j$. For N_i , the root part is controlled by the choice $R_0 \in O_i$, and the $A_i \setminus R$ part agrees with S_i . The only new contribution comes from the tail $A_j \setminus R$, which is disjoint from A_i and lies outside $H_i = A_i$. Hence $P_{A_j \setminus R} = P_{A_j \setminus R} P_{\Gamma \setminus H_i}$, and for each k ,

$$\|S_j P_{A_j \setminus R} x_{i,k}\|_p \leq \|S_j\| K^2 \|P_{\Gamma \setminus H_i} x_{i,k}\| < \varepsilon_i / 8.$$

Hence the defining inequalities of N_i hold. The argument for N_j is identical. Thus $S \in U_i \cap U_j$, contradicting pairwise disjointness. This proves CCC. $\square$

Corollary 2. *Let $0 < p \leq 1$, let $p \leq r < \infty$, and let $(E_\gamma)_{\gamma \in \Gamma}$ be any family of separable p -natural quasi-Banach spaces. If*

$$E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_r \quad \text{or} \quad E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{c_0},$$

then $C_{\text{ph}}^b(B_{E^\Delta}, L_p[0, 1])$ satisfies the countable chain condition. In particular, for every set Γ , the source conjecture holds for $E = \ell_r(\Gamma)$ whenever $p \leq r < \infty$, and for $E = c_0(\Gamma)$.

Proof. The p -naturality of these sums follows from the coordinatewise embedding of the blocks into the corresponding sums of their witnessing p -convex p -Banach lattices. In the finite r case, the p -triangle inequality and p -convexity follow from Minkowski's inequality in $\ell_{r/p}$, since $r/p \geq 1$. The c_0 case is immediate from taking suprema coordinatewise. The usual coordinate projections form a contractive separable Boolean coordinate decomposition, so the conclusion follows from Theorem 2. $\square$

Verification notes

The proof does not settle the full conjecture. Its structural assumption is that E has a bounded separable Boolean coordinate decomposition. The source-paper obstruction remains for arbitrary subspaces of products: topological CCC is not hereditary, so a general p -natural space embedded in $\ell_\infty(J, L_p[0, 1])$ need not be covered by this argument.

The bounded literature check for this packet consisted of local index searches for arXiv id 2512.05273, the title, p -natural, free quasi-Banach lattice, C_{ph}^b , and countable chain condition, together with web searches for the exact title and CCC conjecture phrases. The search found the source arXiv page but no later answer to the Section 7.2 conjecture.

Human-review recommendation

Review Theorems 1 and 2 as likely valid partial positive results. For Theorem 1, the main technical point is Lemma 3, especially the continuity of the inverse homeomorphism for elements of the p -sum with infinite countable support. For Theorem 2, the main point is the finite-block density and root-gluing argument after the operators have been shrunk inside homogeneous cones.

References

- [1] A. Salguero-Alarcon, P. Tradacete, and N. Trejo-Arroyo, *Free quasi-Banach lattices*, arXiv:2512.05273, 2025.
- [2] R. Engelking, *General Topology*, Heldermann Verlag, 1989.